

Binomial distribution A discrete random variable X is said to follow binomial distribution if its probability mass function is

$$P(X=x) = {}^n C_x p^x q^{n-x}, \quad x=0, 1, 2, \dots, n, \quad q=1-p.$$

Moment generating function $M_X(t) = E(e^{tx})$

$$= \sum_{x=0}^n e^{tx} p(X=x)$$

$$= \sum_{x=0}^n e^{tx} {}^n C_x p^x q^{n-x} = \sum_{x=0}^n {}^n C_x (pe^t)^x q^{n-x} = (pe^t + q)^n$$

Mean of Binomial distribution $E(X) = \left[\frac{d}{dt} M_X(t) \right]_{t=0}$

$$= \left[n(pe^t + q)^{n-1} \cdot pe^t \right]_{t=0}$$

$$= [n(p+q)^{n-1} \cdot p]$$

$$= np \quad (\text{since } q=1-p)$$

$$E(x^2) = \left[\frac{d^2}{dt^2} M_x(t) \right]_{t=0} = \left\{ \frac{d}{dt} \left[n(pe^t + q)^{n-1} pe^t \right] \right\}_{t=0}$$

$$= \left[n(n-1)(pe^t + q)^{n-2} p^2 e^{2t} + n(pe^t + q)^{n-1} \cdot pe^t \right]_{t=0}$$

$$= n(n-1)p^2 + np$$

$$\text{Var}(X) = E(x^2) - [E(X)]^2 = n(n-1)p^2 + np - n^2p^2 = np - np^2$$

$$= np(1-p)$$

$$\therefore \boxed{\text{Var}(X) = npq}$$

Remarks:- 1) A binomial experiment consists of n independent trials. If we consider N sets of n independent trials then the number of times we get x success is

$$N \cdot {}^nC_x p^x q^{n-x}$$

Problem: For a binomial distribution with parameters $n=5$, $p=0.3$. Find the probabilities of getting

- at least 3 successes
- at most 3 Successes
- exactly 3 failures.

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Soln:- Let X be a binomial random variable with parameters n and p . Then

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$\begin{aligned} \text{(i)} \quad P(X \geq 3) &= P(X=3) + P(X=4) + P(X=5) \\ &= {}^5C_3 (0.3)^3 (0.7)^2 + {}^5C_4 (0.3)^4 (0.7) + {}^5C_5 (0.3)^5 (0.7)^0 \\ &= 0.1631. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= {}^5C_0 (0.3)^0 (0.7)^5 + {}^5C_1 (0.3)^1 (0.7)^4 + {}^5C_2 (0.3)^2 (0.7)^3 \\ &\quad + {}^5C_3 (0.3)^3 (0.7)^2 \\ &= 0.9692. \end{aligned}$$

$$\text{(iii)} \quad P(X=3 \text{ failures}) = P(X=2 \text{ successes}) = {}^5C_2 (0.3)^2 (0.7)^3 = 0.3087$$

Problem: If the chance of running a bus service according to schedule is

0.8, Calculate the probability on a day schedule with 10 services:

- exactly one is late
- at least one is late.

Soln: - Let X denote the number of buses which are late in their schedule. Hence X follows the binomial distribution.

Here $n=10$; $p=0.2$ $q=0.8$

$$\therefore p(X=x) = {}^{10}C_x (0.2)^x (0.8)^{10-x}$$

$$p(X=1) = {}^{10}C_1 (0.2)^1 (0.8)^9 = 0.2684$$

$$p(X \geq 1) = 1 - p(X < 1) = 1 - p(X=0)$$

$$= 1 - {}^{10}C_0 (0.2)^0 (0.8)^{10} = 0.8926$$

Pb:-3 The mean of a binomial distribution is 5 and standard deviation is 2. Find the distribution.

Soln:-

$$\begin{aligned} \text{Given } E(X) &= 5 & ; & \quad \sqrt{\text{var}(X)} = 2 \\ np &= 5 & ; & \quad \sqrt{npq} = 2 \Rightarrow npq = 4 \end{aligned}$$

$$\begin{aligned} \text{Now, } \frac{npq}{np} &= \frac{4}{5} \Rightarrow \boxed{q = 4/5} \\ \text{Since } p &= 1 - q \Rightarrow \boxed{p = 1/5} \end{aligned}$$

$$\text{from } np = 5 \Rightarrow \boxed{n = 25}$$

$\therefore$ The probability distribution is $P(X=x) = {}^{25}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{25-x}$

$$x = 0, 1, 2, \dots, 25$$

Problem: 4 coins are tossed and number of heads noted. The experiment is repeated 200 times and the following distribution is obtained.

X: Number of heads	0	1	2	3	4
f: frequencies	62	85	40	11	2

Find the binomial distribution and expected frequency distribution.

Soln:-

Here $n = 4$; $N = 200$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{206}{200} = 1.03$$

$$\therefore np = 1.03$$

$$p = \frac{1.03}{4} \Rightarrow \boxed{p = 0.257}$$

$$q = 1 - 0.257 = 0.743$$

$\therefore$ The required Binomial distribution is

$$P(X = x) = 200 \times {}^4C_x (0.257)^x (0.743)^{4-x}$$